GlucoTwin

A local synthetic digital twin for a two-hour glucose forecast

- Digital Twin Challenge 2026 | Team: [Add team name] | College: [Add institution]

- A proof of concept for exploring patient context plus a simulated sensor stream.

- All people and observations are fictional. Not for clinical use.

The use case

Could patient-specific context help estimate glucose two hours ahead?

- Forecast target is glucose at current timestamp + 120 minutes.

- A future governed research workflow could inspect trends and forecasting concepts.

- This prototype is not a medical device and does not support diagnosis or treatment.

Data and model

Seeded generation makes the complete demo reproducible

- 60 synthetic EHR-style profiles; 120,960 simulated readings at 15-minute intervals.

- Features: glucose, heart rate, activity, sleep, time-of-day, age, BMI, HbA1c and duration.

- Chronological holdout: earlier timestamps train, later timestamps test; compare with last-value persistence.

Model MAE	Baseline MAE	Test rows
5.048 mg/dL	8.567 mg/dL	24,060

Live demo

Streamlit dashboard and virtual patient state

- Choose a synthetic patient and inspect fictional profile fields.

- Advance the simulated sensor stream one 15-minute record at a time.

- Read the current simulated value separately from the predicted value at +2 hours.

- Metrics are actual checked-in synthetic evaluation, not clinical validation.

Limits and next steps

Responsible interpretation comes first

- Synthetic physiology is simplified and does not represent device noise or patient diversity.

- No external, prospective, subgroup, or clinical validation has been performed.

- Before clinical research: governance, consent, data quality, external validation, safety, and fairness review.

- Demo video: [Add a real recording link after it exists].